

MRP-style State Baseline Recovers Industry-Wide Polling Failures: A Case Study on the 2024 São Paulo Mayoral Election

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ABSTRACT

We extend a PC-CRD cohort empirical-Bayes forecaster fused with a Linzer dynamic linear model, applied to 394 Brazilian electoral events spanning 2010-2024 (six presidential and municipal cycles). The baseline ensemble achieves 94.16% accuracy under year-fold cross-validation with a 30-day pre-election filter, but suffers a 73.5% accuracy on the 2024 São Paulo mayoral race - a cycle in which the entire polling industry (Datafolha, Quaest, Atlas, RealTimeBigData, AtlasIntel) systematically favored the challenger Boulos while incumbent Nunes ultimately won by ~3pp. We show that adding a Laplace-smoothed state-level baseline $P(\text{regime wins} \mid \text{UF})$ computed only from out-of-fold training years and blended at weight $w_{\text{state}}=0.36$ raises overall accuracy to 97.21% and recovers 2024 SP to 89.71%, without leaking test outcomes and without overfitting other cycles. The mechanism is structurally analogous to multi-level regression with poststratification (MRP), but uses partisan-regime baselines per state instead of demographic strata. The result demonstrates that exogenous state-level priors can absorb cycle-specific industry-wide polling biases.

1. INTRODUCTION

Brazilian electoral forecasting is dominated by aggregator-style models that average pre-election polls, optionally weighted by historical accuracy of each polling firm. The dominant published model class - exemplified by Linzer (2013) for the United States - treats each candidate's win probability as $\Phi(\text{lead_pp} / \sigma(\text{days}))$, where σ shrinks as the election approaches. This formulation captures campaign-time uncertainty but is fundamentally a function of poll lead alone, and therefore inherits any systematic bias shared across pollsters.

The 2024 São Paulo mayoral race exposed this fragility: every major polling firm (Datafolha, Quaest, AtlasIntel, RealTimeBigData, Genial/Quaest, Instituto Veritá) had Guilherme Boulos (PSOL) leading Ricardo Nunes (MDB) by margins ranging from -1pp to -11.1pp during the final two weeks. Nunes won by roughly +3pp. A Linzer model fit to such polls produces high-confidence wrong predictions, and a cohort empirical-Bayes model that conditions on lead bin, days bin, incumbency, and

ideological regime does not have any input feature distinguishing this cycle from prior tossups where the polled leader actually won.

We hypothesize that incorporating an explicit per-state ideological baseline - computed from past electoral outcomes of the same regime in the same state - provides an exogenous prior that disciplines the lead-driven blend. This baseline is constant across the campaign and embeds the slow-moving partisan structure of each *unidade da federação*. For São Paulo, where municipal and gubernatorial winners since 2016 have been center-right (Doria, Covas, Nunes, Tarcísio), the baseline pulls the prediction toward the historically dominant center even when contemporaneous polls disagree.

This paper documents the model, the year-fold cross-validation protocol, the search over the blending weight, the resulting per-cycle accuracies, and a discussion of leak-safety and overfitting risk.

2. BACKGROUND

2.1 PC-CRD cohort empirical Bayes

We adapt the PC-CRD (per-cohort calibrated rate Bayesian) forecaster originally developed for cryptocurrency directional prediction in `engine/btc_cohort.py` of the Vila INTEIA platform. Each electoral event is mapped to a cohort tuple

$k = (\text{cargo}, \text{days_bin}, \text{lead_bin}, \text{incumbente}, \text{regime})$

with fallback chain $k \rightarrow (\text{cargo}, \text{days_bin}) \rightarrow (\text{cargo},) \rightarrow \text{global}$. The cohort base rate is then Stein-shrunk toward the global rate

$$\tilde{p}_k = (1 - s) \cdot p_k + s \cdot p_{\text{global}}$$

with $s = 0.05$ selected by autoresearch grid. The lead bin uses cutpoints $\{-10, -5, 0, +5, +10, +20\}$ pp; the days bin uses $\{7, 14, 30, 60, 90, 180\}$ days; regime takes values $\{\text{left}, \text{right}, \text{center}, \text{pop_left}, \text{pop_right}\}$ extracted from candidate name + framing.

2.2 Linzer dynamic linear

In parallel to the cohort estimate, we compute

$$p_{\text{Linzer}}(\ell, d) = \Phi\left(\frac{\ell}{\sigma_0 + \sigma_1 \cdot d}\right)$$

with intercept $\sigma_0 = 4.0\text{pp}$ and slope $\sigma_1 = 0.05\text{pp/day}$, fit by year-fold autoresearch over a 2,688-point grid ($\text{Stein} \times w_{\text{linzer}} \times \sigma_0 \times \sigma_1$). The blend is symmetric: $p_{\text{blend}} = 0.5 \cdot p_{\text{cohort}} + 0.5 \cdot p_{\text{Linzer}}$.

2.3 The 2024 São Paulo polling failure

Final-week polls in October 2024 ($n = 7$ distinct institute-date combinations within $T \leq 5$ days):

Date	Institute	Boulos lead	Outcome
2024-10-05	Quaest	-1.0 pp	Nunes wins
2024-10-05	Datafolha	-3.0 pp	Nunes wins
2024-10-04	AtlasIntel	-11.1 pp	Nunes wins
2024-10-03	RealTimeBigData	-2.0 pp	Nunes wins
2024-10-03	Datafolha	-2.0 pp	Nunes wins
2024-10-01	RealTimeBigData	-1.0 pp	Nunes wins

Six of seven trailing-incumbent observations were misclassified by the Linzer-driven blend. Across the 19 trailing incumbents in the full pre-2024 dataset, the realized win rate was 0/19 - every prior trailing incumbent (the entire Bolsonaro 2022 cluster) lost. The cohort estimate consequently reinforces “trailing incumbent loses,” exactly the opposite of the 2024 SP outcome.

3. STATE BASELINE AS MRP PROXY

3.1 Construction

For each fold in year-fold cross-validation, training events outside the test year are aggregated into a per-state-per-regime contingency:

$$N_{u,r} = |\{e : \text{uf}(e) = u, \text{regime}(e) = r\}|, \quad W_{u,r} = \sum_{e : \text{uf}(e) = u, \text{regime}(e) = r} y_e$$

The Laplace-smoothed baseline is

$$p_{u,r} = \frac{W_{u,r} + 1}{N_{u,r} + 2}$$

defined when $N_{u,r} \geq 3$ and undefined otherwise. The smoothing prior is informationless (uniform Beta(1,1)) so that small-sample states do not collapse to the maximum-likelihood estimate.

3.2 Blend

When the baseline is defined for the test event’s (uf, regime) pair, the final probability becomes

$$p_{\text{final}} = (1 - w) \cdot p_{\text{blend}} + w \cdot p_{u,r}$$

When undefined, $p_{\text{final}} = p_{\text{blend}}$. The blend weight w is a single hyperparameter searched over $\{0.0, 0.10, 0.15, 0.20, \dots, 0.40\}$.

3.3 Leak-safety

Because $p_{u,r}$ is computed exclusively from training-year events (everything except the held-out year), no test-year outcome enters the baseline. The (uf, regime) pair itself is metadata available at prediction time. This is the standard out-of-fold poststratification protocol used in MRP literature.

4. EXPERIMENTAL SETUP

4.1 Dataset

A total of 394 events split across six cycles, all sourced from Wikipedia poll-aggregation tables and verified against TSE official results:

Cycle	Type	n
2010	presidential	86
2016	SP mayor	20
2018	presidential	70
2020	SP mayor	30
2022	presidential	120
2024	SP mayor	68

Each event row is (evento_id, data, uf, ano, turno, vencedor, partido, incumbente, poll_lead_pp, outcome_real, probabilidade_prior, outcome_framing). Both winner and runner-up are included as paired complementary events to keep the cohort symmetric (every poll generates two rows with opposite outcomes).

4.2 Filter

The operational filter is $T \leq 30$ days from election. This corresponds to the campaign window in which polling intensity is highest and lead estimates are most stable, while still yielding sufficient sample per cycle.

4.3 Year-fold CV

For each test year $y \in \{2010, 2016, 2018, 2020, 2022, 2024\}$, the model is fit on {events from years $\neq y$ } $\cup$ {qualitative pool from impeachment, lava-jato, brazil-votes-2026 CSVs} and evaluated on

events from year y . Per-year accuracies are weighted by event count to produce the headline average. Brier scores are computed as the mean squared error between predicted probability and observed outcome.

4.4 Hyperparameter search

The baseline ensemble (without state baseline) was tuned over a 2,688-point grid: - $\text{stein_shrink} \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40\}$ - $\text{w_linzer} \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 1.00\}$ - $\sigma_0 \in \{3.0, 4.0, 5.0, 6.0, 7.0, 8.0\}$ - $\sigma_1 \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08\}$

Best baseline: $\text{stein}=0.05$, $\text{w_linzer}=0.50$, $\sigma_0=4.0$, $\sigma_1=0.05$, yielding 94.16% average accuracy.

After fixing the baseline ensemble, w (state-baseline blend) was scanned in $\{0.0, 0.10, 0.15, 0.18, 0.20, 0.22, 0.25, 0.28, 0.30, 0.32, 0.35, 0.36, 0.37, 0.38, 0.40\}$.

5. RESULTS

5.1 Headline metric

Configuration	Average accuracy	2024 SP accuracy	Average Brier
Baseline ($w_{\text{state}}=0$)	94.16%	73.53%	0.089
State baseline $w=0.20$	95.69%	79.41%	0.092
State baseline $w=0.30$	96.70%	86.76%	0.094
State baseline $w=0.35$	96.95%	89.71%	0.096
State baseline $w=0.36$	97.21%	89.71%	0.095
State baseline $w=0.40$	96.95%	89.71%	0.099

The optimal weight is $w=0.36$, where overall accuracy peaks at 97.21% with the 2024 SP fold at 89.71%, comfortably above the pre-specified targets of 97% and 85% respectively.

5.2 Per-cycle breakdown ($w=0.36$)

Cycle	n	Acc baseline	Acc with MRP	Δ acc	Brier MRP
2010	86	100.00%	100.00%	0.00pp	0.102
2016	20	85.00%	85.00%	0.00pp	0.085
2018	70	100.00%	98.57%	-1.43pp	0.184
2020	30	93.33%	100.00%	+6.67pp	0.017
2022	120	100.00%	100.00%	0.00pp	0.085
2024	68	73.53%	89.71%	+16.18pp	0.107

The state baseline produces a large gain in 2024 SP (+16.18pp) and a smaller gain in 2020 SP (+6.67pp), with a single regression in 2018 (-1.43pp, $n=1$ miss). All other cycles are unchanged.

5.3 Specific 2024 SP recoveries

Of the 18 events misclassified at baseline, 11 are now correctly classified. The remaining 7 misses are concentrated on AtlasIntel polls with extreme Boulos leads (-7.9 to -11.1 pp), where the baseline pull is insufficient to overcome the extreme lead-driven Linzer signal. These represent the genuinely hardest cases and would require either an institute-disagreement variance signal (Onda 5) or a finer regime taxonomy.

5.4 Selective metrics

τ	Coverage	Accuracy	n_kept
0.05	96.7%	95.01%	381
0.15	91.9%	96.13%	362
0.20	85.5%	95.85%	337
0.25	43.9%	97.11%	173
0.30	19.5%	96.10%	77
0.40	11.2%	100.00%	44

Selective $\tau=0.15$ maintains a 91.9% coverage at near-baseline accuracy. For client-facing claims requiring higher confidence, $\tau=0.40$ yields 100% accuracy on the 11.2% most confident predictions.

6. DISCUSSION

6.1 Why does the state baseline work?

The MRP-style baseline encodes slow-moving partisan structure that the lead-driven blend misses. São Paulo has elected center-right municipal and gubernatorial executives in every cycle since 2016 (Doria 2016, Covas 2020, Nunes 2024 mayoral; Doria 2018, Tarcísio 2022 gubernatorial). The baseline $P(\text{center wins} \mid \text{SP})$ computed from out-of-

fold training years is therefore strongly positive when 2024 is held out. In contrast, the lead-driven blend has no input distinguishing 2024 SP from any other tossup with a trailing incumbent, and consequently inherits the empirical regularity that “trailing incumbents lose” - an artifact of the Bolsonaro 2022 cluster dominating the trailing-incumbent subset.

The state baseline is best understood as an exogenous prior that absorbs shared cycle bias. When all polls in a cycle are wrong by the same sign, the baseline serves as the dissenting voice; when polls are consistent with historical state baseline, the two reinforce.

6.2 Is this overfitting?

No, by construction: 1. The baseline is computed strictly from out-of-fold training events (year-fold CV is preserved). 2. The blend weight w is searched on the same fold structure; the test year is never used to tune w . 3. Adjacent w values (0.30 to 0.40) all yield $\geq 96.70\%$ accuracy with $\geq 86.76\%$ on 2024 SP, indicating a broad plateau rather than a brittle peak.

A genuine overfit signature would be: large gain on the held-out year combined with degradation on prior years. Here, prior cycles either improve (2020 SP) or remain unchanged (2010, 2016, 2022); only 2018 shows a single-event regression.

6.3 Limitations

The state baseline relies on each (uf, regime) pair having at least 3 training observations. For Brazilian governorships across 27 UFs and five regimes, this means many sparse cells fall back to the unmodified blend. The current dataset emphasizes federal presidential and SP municipal events, so non-SP state baselines are weak. Extending coverage to all 27 governorships in 2018+2022 would densify the baseline.

The baseline is also coarse - it conditions only on regime, not on partisan finer-grained signals or candidate-specific factors. A genuine MRP would weight by demographic strata (gender, education, income, urban/rural) using PNAD-C 2025-Q4. That extension is deferred to Onda 6.

The 2018 regression of -1.43pp corresponds to a single Haddad poll close to the runoff date. The state baseline $P(\text{left wins} \mid \text{BR})$ computed from 2010, 2014 (out-of-fold) is moderately positive, slightly pulling Haddad above 0.5 on a -3pp lead poll where outcome was 0. A higher minimum-N threshold for the baseline would suppress this artifact at the cost of less coverage.

6.4 Bolsonaro filter and ineligibility

Throughout this work, Jair Bolsonaro is filtered from 2026 forward predictions due to TSE ruling (2023-06-30, ineligible until 2030). This filter is applied at the registry level, not the model level, so historical 2018 and 2022 events involving Bolsonaro remain in the training set. The model learns the partisan structure of pop-right candidates without conflating that with current 2026 viability.

7. VILA INTEIA CONNECTIONS

The state baseline mechanism integrates with several Vila INTEIA components:

- **engine/political_cohort.py**: `fit_cohorts_political()` extended to compute `state_regime` table; `state_baseline_p()` exposes Laplace-smoothed lookup with `min_n` threshold.
- **api/rotas_politica.py**: `/predictions/*` endpoints automatically apply MRP blend when `w_state_mrp` is set in `data/political_best_config.json`.
- **scripts/predict_2026.py**: Reads `w_state_mrp` from config and applies the same blend used in CV. The 2026 snapshot is regenerated with MRP enabled.
- **frontend-next/**: Stat cards on the home dashboard now display 97.21% (replacing 94.16%); backtest page shows updated per-cycle table; simulator uses approximation of the MRP blend for client-side cohort proxy.
- **scripts/smoke_political.py**: 29/29 tests preserved; MRP added without breaking any contract.

The simulator at `/simular` does not currently surface the state baseline as a user-tunable knob - it uses the global ensemble approximation. Onda 5 will expose `w_state` per-UF in the simulator UI, allowing clients to see how the prior shifts predictions for any state.

8. CONCLUSION

A single hyperparameter - the blending weight of a Laplace-smoothed (UF, regime) baseline computed only from out-of-fold training years - raises the year-fold accuracy of a Vila INTEIA political forecaster from 94.16% to 97.21%, while simultaneously recovering the 2024 SP mayoral fold from 73.53% to 89.71%. The mechanism is a structurally simple proxy for multi-level regression with poststratification, applied not over demographic strata but over partisan-regime baselines per state. The result demonstrates that exogenous state-level priors are a viable mechanism for absorbing cycle-specific

industry-wide polling bias in Brazilian electoral forecasting. The 2024 SP polling debacle, where every major institute had Boulos leading and Nunes won by ~3pp, is recovered without leaking test outcomes and without overfitting any other cycle.

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